

Introduction to Lie Group Theory

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Chapter 1

Matrix Groups

1.1 Groups of Matrices

1.2 Complex Matrices as Real Matrices

1.3 Matrix Groups of Normed Vector Spaces

1.4 Matrix Exponential and Logarithm

1.5 One Parameter Subgroups in Matrix Groups

Chapter 2

Tangent Spaces and Lie Algebras

2.1 Lie Algebras

2.2 Curves and Tangent Spaces

2.3 Lie Algebras of Matrix Groups

2.4 $SO(3)$ and $SU(2)$

2.5 Complexification of Real Lie Algebra

Chapter 3

Lie Groups

- 3.1 Smooth Manifolds**
- 3.2 Tangent Spaces and Derivatives**
- 3.3 Lie Groups**
- 3.4 Matrix Groups are Lie Groups**
- 3.5 Not All Lie Groups are Matrix Groups**

References

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